

HAT-P-55b

R-0056

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-55_250627 · created 2026-07-10T06:21:50+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460853.86874 +/- 0.00053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1625 +/- 0.0022
Transit depth (Rp/Rs)^2	2.64 +/- 0.073 [%]
Orbital Inclination (inc)	84.381 +/- 0.088
Airmass coefficient 1 (a1)	0.952 +/- 0.016
Airmass coefficient 2 (a2)	0.0551 +/- 0.0088
Scatter in the residuals of the lightcurve fit is	1.69 %
Best Comparison Star	#2 - [386, 233]
Optimal Aperture	2.5
Optimal Annulus	7.12
Transit Duration (day)	0.0694 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1625 ± 0.0022 vs 0.1202 (deviation 2.94σ)
Duration fit vs published	0.0694 d vs 0.1225 d (deviation 7.37σ)
Mid-time O–C	23.9 min
Depth SNR	11.3
Residual scatter	1.69 %

Light curve

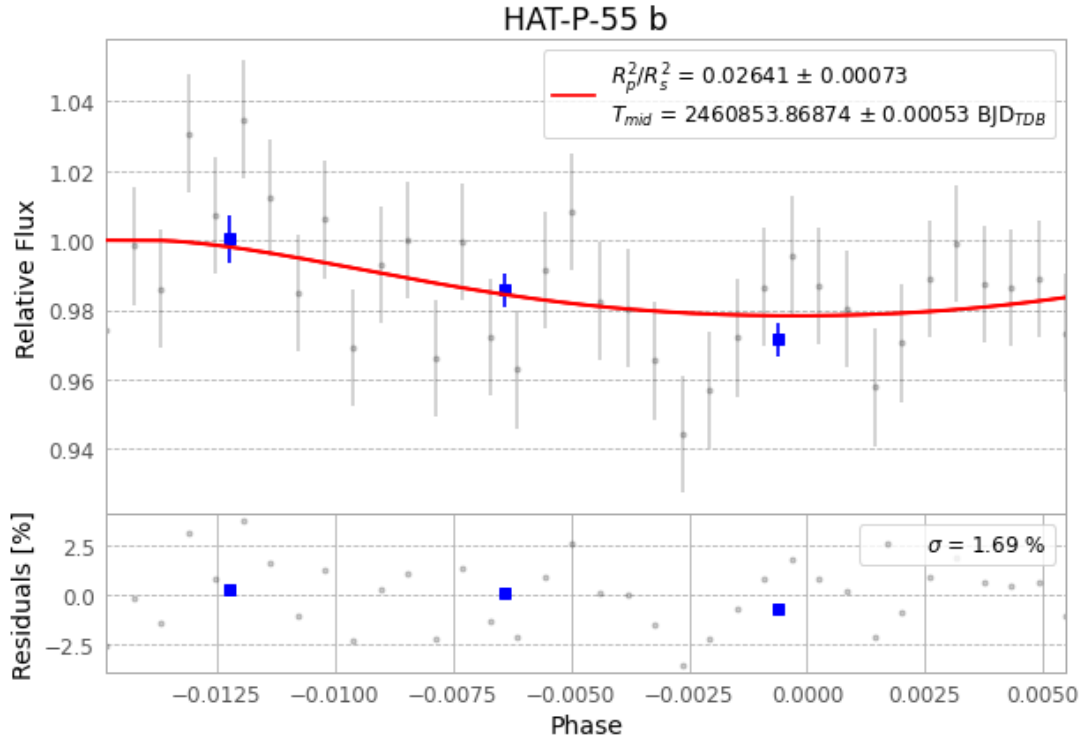


Figure 1 — FinalLightCurve_HAT-P-55 b_2025-06-27.png (SHA-256 7340f7c37b23e7da...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2ef640e5724f6de2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

36 FITS frames (24 MB), HATP-55250627072715.FITS → HATP-55250627091215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-55-250627.log (8868813ff79b...), FinalLightCurve_HAT-P-55 b_2025-06-27.png (7340f7c37b23...), FinalLightCurve_HAT-P-55 b_2025-06-27.csv (cb705a9fea8a...)

Compute resources

Wall time 201.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.